

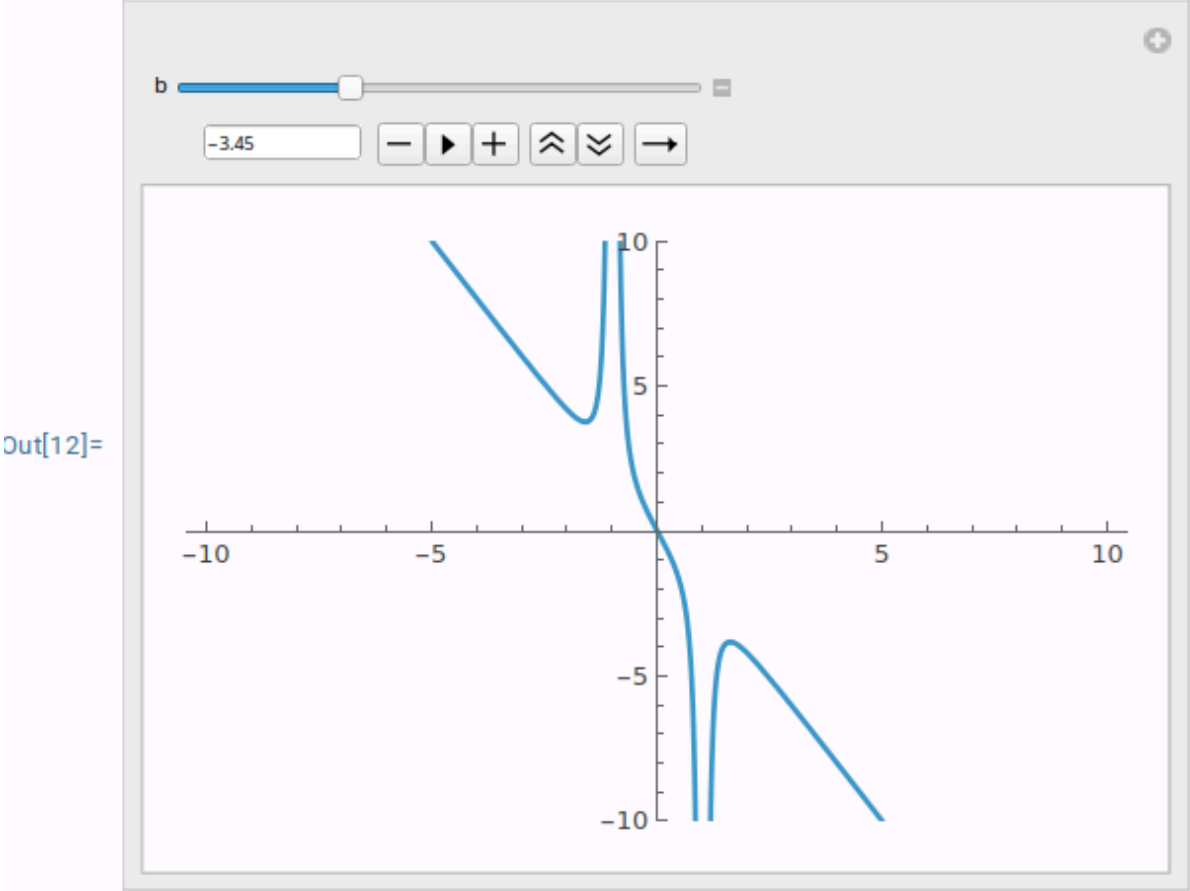
SAC 1 - inflection point conditions Q

Given $k(x) = \frac{x^4+bx^2+2}{1-x^2}$, find values of b such that the graph of $k(x)$ has no inflection points?

- Generally to find inflection points, we want to find values of x s.t $k''(x) = 0$ s.t x is a turning point of the graph of $k'(x)$. To test whether x is a t.p of k' , we can use a sign test or check if $k'''(x)$ is non-zero.
- Playing around with a manipulated plot, what should we expect from our final answers?

```
In[8]:= k' [x]
Out[8]=  $\frac{2bx + 4x^3}{1-x^2} + \frac{2x(2+bx^2+x^4)}{(1-x^2)^2}$ 

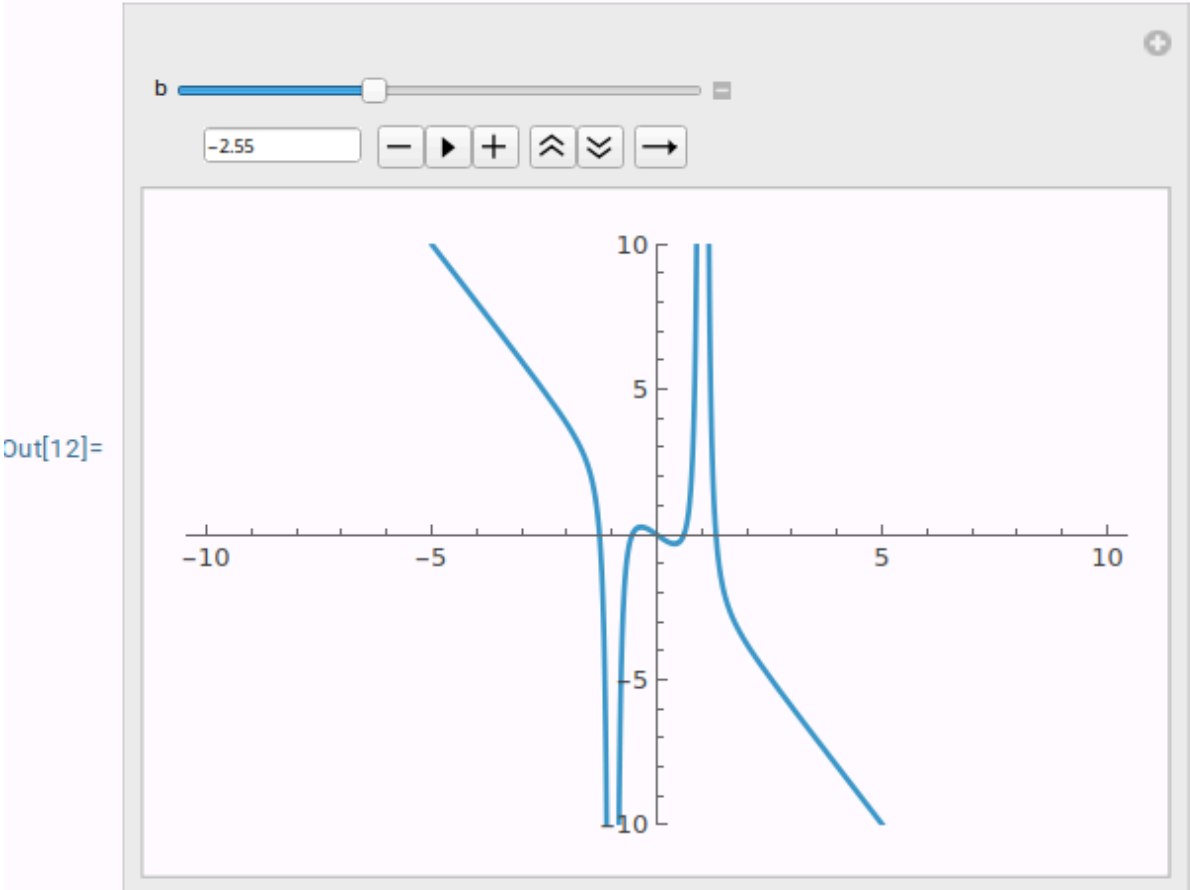
In[12]:= Manipulate[Plot[ $\frac{2bx + 4x^3}{1-x^2} + \frac{2x(2+bx^2+x^4)}{(1-x^2)^2}$ , {x, -10, 10}, PlotRange -> 10], {b, -10, 10}]
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- Seems like for $b < -3$ the graph of k' pretty much stays this shape

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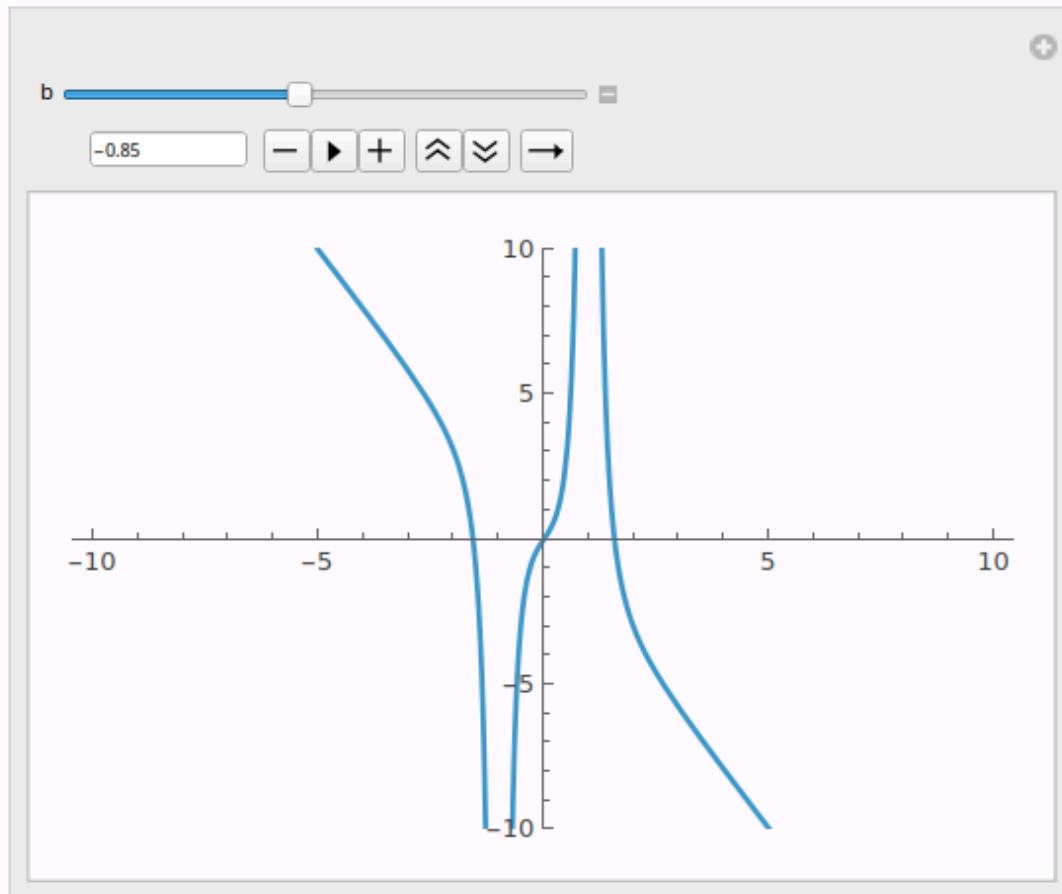
-
- For $-3 < b < -2$ the graph completely changes shape, and there seems to be two tp's near the centre

In[8]:= $k'[x]$

$$\text{Out[8]} = \frac{2bx + 4x^3}{1 - x^2} + \frac{2x(2 + bx^2 + x^4)}{(1 - x^2)^2}$$

In[12]:= $\text{Manipulate}\left[\text{Plot}\left[\frac{2bx + 4x^3}{1 - x^2} + \frac{2x(2 + bx^2 + x^4)}{(1 - x^2)^2}, \{x, -10, 10\}, \text{PlotRange} \rightarrow 10\right], \{b, -10, 10\}\right]$

Out[12]=



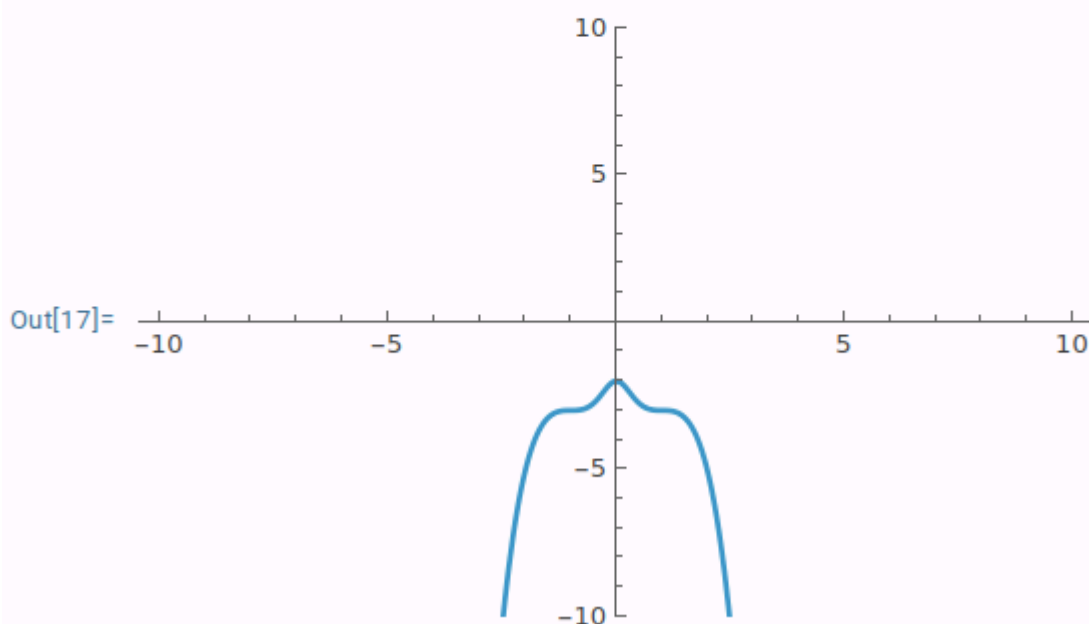
- For $b > -2$ The graph pretty much stays in this shape, with no inflection points!
 - Building cases:
 - Clearly for $b > -2$ there are no tp's of k' hence no inflection points of k
 - for $b \in (-\infty, -3) \cup (-3, -2)$ there seems to exist some tp's of k' and hence inflection points of k
 - Lets check cases
 - $b = -3$: by cancellation of factors in the numerator and denominator, k becomes a parabola, so no inflection points
 - $b = -2$: Verify yourself; no inflection point. Why?
 - So overall, for $b \in \{-3\} \cup [-2, \infty)$ there are no inflection points on the graph of k
- Lets actually show why this is the case:

1. Lets plot $k''(x) = 0$ on a $b - x$ plane (i.e, solve for the parameter and plot the "parameter" graph)

In[16]:= $\text{Solve}[k''[x] == 0, b] // \text{FullSimplify}$

$$\text{Out[16]} = \left\{ \left\{ b \rightarrow -\frac{2 + 12x^2 - 3x^4 + x^6}{1 + 3x^2} \right\} \right\}$$

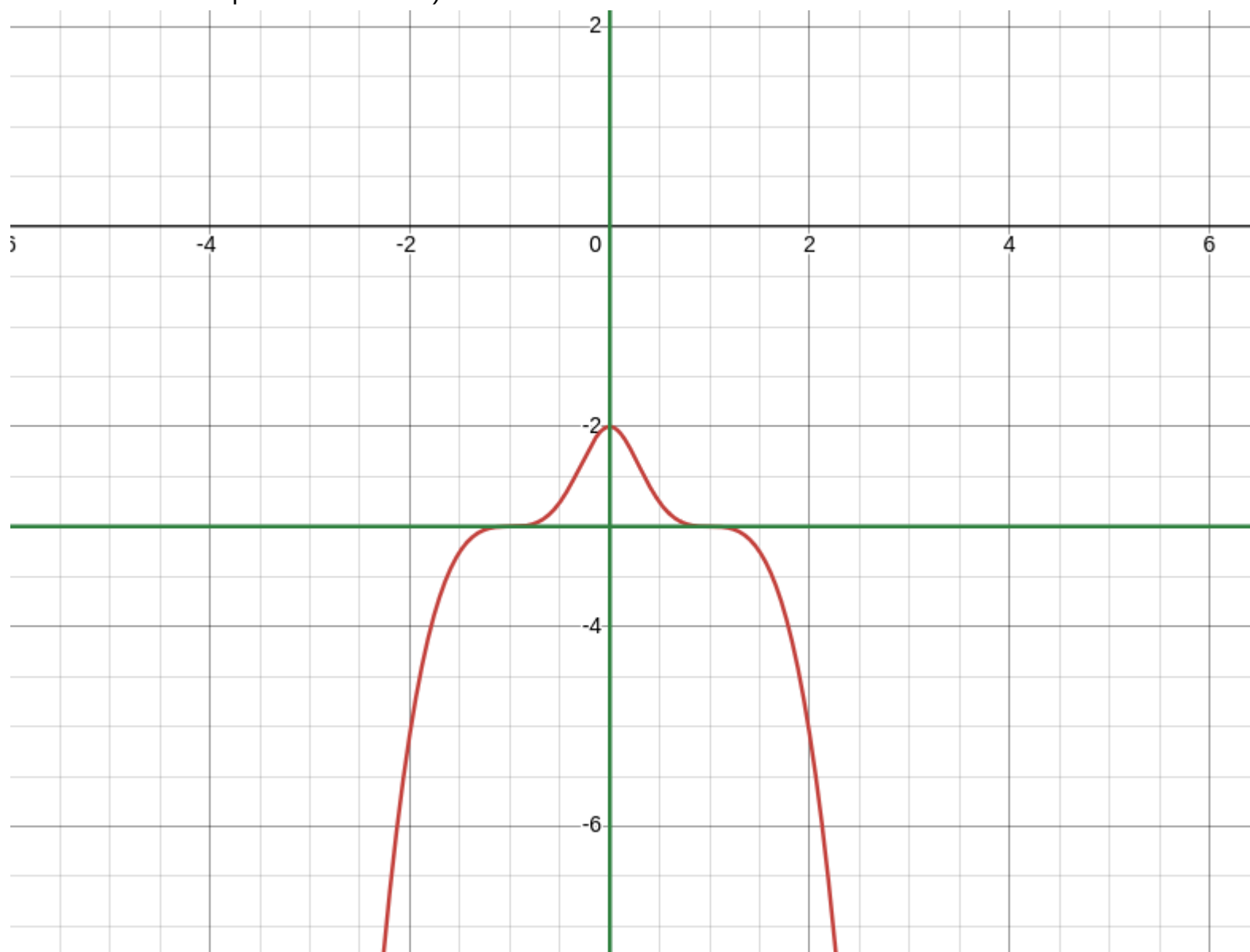
In[17]:= $\text{Plot}\left[-\frac{2 + 12x^2 - 3x^4 + x^6}{1 + 3x^2}, \{x, -10, 10\}, \text{PlotRange} \rightarrow 10\right]$



2. For a given value of $b = \beta$, if $k(x)$ does NOT have an inflection point then one of the following must be true:

1. $k''(x) = 0$ has no solutions (k' graph has no stationary points so they cant have any tps!)
 - $k''(x) = 0$ has no solutions only when $b > -2$. This is when the inflection point in the centre stops being stationary, and becomes a non stationary infl. point.
2. all solutions of $k''(x) = 0$ satisfy $k'''(x) = 0$ (k' graph has stationary points, but they are inflection points NOTE THIS IS ON THE k' GRAPH)

- lets plot all the (b, x) pairs of values for b and x that satisfy $k''(x) = 0$ (i.e the parameter graph) and $k'''(x) = 0$ (also a parameter graph, but for a different equation of interest!)



- The parameter graph from $k'''(x) = 0$ actually gives us two straight lines.
- We want solutions to $k''(x) = 0$ that also satisfy $k'''(x) = 0$. Translating this to refer to our parameter graphs:
 - we want points on the red curve (red points satisfy $k''(x) = 0$) to also lie on the green lines (green points satisfy $k'''(x) = 0$)
 - we want the points in the intersection of the red and green "curves"!
 - quotation marks bc here the green "curve" is just a pair of lines. But in general it can be a more complicated curve
 - These are points corresponding to $b = -3, -2$
 - Why does $b = -2$ work? :)
- Putting the above two points together,
 - No inflection points when $b > -2$
 - No inflection points when $b = -3$ and $b = -2$